Math 241 Sec 5

Faculty of Engineering
Cairo University

Echelon Forms of a Matrix

Properties:

REF

1. **Zero rows** (rows where all entries are zeros), **if exist**, appear at the **bottom**.
2. The **first nonzero entry from the left** in any row is a **1** (called a ***leading 1***).
3. A ***leading 1*** **appears to the right and bottom** of the ***leading 1*** in the preceding row ("echelon" or staircase pattern).

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4. In **a column that contains a *leading 1***, all other entries are zeros.

row echelon from

① ever leader in a row is one

② every lead in the next row is on the right the leader in perved row

$$\begin{bmatrix} 1 & \cdots & & \\ 0 & 1 & & \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$ no

③ the zero row is later row

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

④ reduced

every leading has zeros above and below of leading

transformation

REF

3.Which of the following are in row-echelon form?

row

(b) $\begin{bmatrix} 1 & -7 & 5 & 5 \\ 0 & 1 & 3 & 2 \end{bmatrix}$

Yes

(d) $\begin{bmatrix} 1 & 3 & 0 & 2 & 0 \\ 1 & 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

No

(f) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Yes

h. $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 2 & 1 \end{bmatrix}$

No

j. $\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ **Yes**

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2.Which of the following are in reduced row-echelon form?

(b) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

No

(d) $\begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

~~No~~ Yes

(f) $\begin{bmatrix} 1 & 0 & 3 & 1 \\ 0 & 1 & 2 & 4 \end{bmatrix}$

Yes

1. Find a row echelon form of each of the given matrices. Record the row operations you perform, using the notation for elementary row operations.

(a) $A = \begin{bmatrix} -1 & 2 & -5 \\ 2 & -1 & 6 \\ 1 & 1 & 3 \end{bmatrix}$

T (b) $A = \begin{array}{c} \begin{bmatrix} 3 & 4 & -1 \\ 6 & 7 & -4 \\ 1 & 1 & -1 \\ 5 & 6 & -3 \end{bmatrix} \end{array} \begin{array}{l} R_1 \\ R_2 \\ R_3 \\ R_4 \end{array}$

Sol

$$\xrightarrow{R_3 \leftrightarrow R_1} \begin{bmatrix} 1 & 1 & -1 \\ 6 & 7 & -4 \\ 3 & 4 & -1 \\ 5 & 6 & -3 \end{bmatrix}$$

$$\xrightarrow[-5R_1 + R_4 \to R_4]{\substack{-6R_1 + R_2 \to R_2 \\ -3R_1 + R_3 \to R_3}} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

$$\xrightarrow[-R_2 + R_4 \to R_4]{-R_2 + R_3 \to R_3} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

2. Find a row echelon form of each of the given matrices. Record the row operations you perform, using the notation for elementary row operations.

T (a) $A = \begin{bmatrix} 1 & -1 & 1 & 0 & -3 \\ -1 & 2 & 3 & 1 & 4 \\ 3 & 4 & 1 & 1 & 10 \end{bmatrix}$

$$\xrightarrow[-3R_1+R_3 \to R_3]{R_1+R_2 \to R_2} \begin{bmatrix} 1 & -1 & 1 & 0 & -3 \\ 0 & 1 & 4 & 1 & 1 \\ 0 & 7 & -2 & 1 & 19 \end{bmatrix}$$

$$\xrightarrow{-7R_2+R_3 \to R_3} \begin{bmatrix} 1 & -1 & 1 & 0 & -3 \\ 0 & 1 & 4 & 1 & 1 \\ 0 & 0 & -30 & -6 & 12 \end{bmatrix}$$

$$\xrightarrow{\frac{-1}{30}R_3 \to R_3} \begin{bmatrix} 1 & -1 & 1 & 0 & -3 \\ 0 & 1 & 4 & 1 & 1 \\ 0 & 0 & 1 & \frac{1}{5} & \frac{-2}{5} \end{bmatrix}$$

3. Each of the given matrices is in row echelon form. Determine its reduced row echelon form. Record the row operations you perform, using the notation for elementary row operations.

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(a) $A = \begin{bmatrix} 1 & 3 & 5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$

(b) $A = \begin{bmatrix} 1 & 3 & 5 & -4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Sol

$$\xrightarrow{-5R_2+R_1 \to R_1} \begin{bmatrix} 1 & 3 & 0 & -14 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow[-2R_3+R_2 \to R_2]{14R_3+R_1 \to R_1} \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

RREF

4. Each of the given matrices is in row echelon form. Determine its reduced row echelon form. Record the row operations you perform, using the notation for elementary row operations.

T (a) $A = \begin{bmatrix} 1 & 0 & 6 & -3 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$-6R_3 + R_1 \to R_1$

$-3R_3 + R_2 \to R_2$

$$\begin{bmatrix} 1 & 0 & 0 & -15 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

6. Find the reduced row echelon form of each of the given matrices. Record the row operations you perform, using the notation for elementary row operations.

(a) $A = \begin{bmatrix} 0 & 3 & -4 \\ 2 & -1 & 6 \\ -1 & 2 & -5 \end{bmatrix}$

T (b) $A = \begin{bmatrix} 1 & 3 & 5 & -2 \\ 1 & 4 & 6 & -2 \\ -1 & -1 & -3 & 2 \end{bmatrix}$

Sol

$$\xrightarrow[R_1 + R_3 \to R_3]{-R_1 + R_2 \to R_2} \begin{bmatrix} 1 & 3 & 5 & -2 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 2 & 0 \end{bmatrix}$$

$$\xrightarrow[-2R_2 + R_3 \to R_3]{-3R_2 + R_1 \to R_1} \begin{bmatrix} 1 & 0 & 2 & -2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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2.2 Exercises

1. Each of the given linear systems is in row echelon form. Solve the system.

T (a) $\begin{aligned} x + 3y + z &= 6 \\ y + z &= 5 \\ z &= 4 \end{aligned}$ (b) $\begin{aligned} x - 4y + 5z + w &= 0 \\ z - 2w &= 4 \\ w &= 1 \end{aligned}$

2. Each of the given linear systems is in row echelon form. Solve the system.

(a) $\begin{aligned} x + 2y - z + 3w &= 4 \\ w &= 5 \end{aligned}$ T (b) $\begin{aligned} x - 3y + 4z &= 0 \\ y + 2z &= 0 \\ z &= 1 \end{aligned}$

T (a) $\begin{aligned} x + 3y + z &= 6 \\ y + z &= 5 \\ z &= 4 \end{aligned}$

sol $\boxed{z = 4}$

eq(2) $\rightarrow$ $y + 4 = 5 \rightarrow \boxed{y = 1}$

eq(1) $\rightarrow$ $x + 3 + 4 = 6$

$\boxed{x = -1}$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix}$$

T (b) $x - 3y + 4z = 0$

$y + 2z = 0$

$z = 1$

So

$\xrightarrow{eq(3)} z = 1$

$\xrightarrow{eq(2)} y + 2 = 0 \rightarrow \boxed{y = -2}$

$\xrightarrow{eq(1)} x + 6 + 4 = 0$

$\boxed{x = -10}$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -10 \\ -2 \\ 1 \end{bmatrix}.$$

4. Each of the given linear systems is in reduced row echelon form. Solve the system.

(a) $\begin{aligned} x \quad - 4z &= 5 \\ y + 2z &= 2 \end{aligned}$ (b) $\begin{aligned} x \qquad\qquad &= 1 \\ y \qquad &= 2 \\ z - 3w &= 4 \end{aligned}$

Sol

(a) $\begin{aligned} x \quad - 4z &= 5 \\ y + 2z &= 2 \end{aligned}$

eq(2) $\boxed{y = 2 - 2z}$

eq(1) $\boxed{x = 5 + 4z}$

let $\boxed{z = t}$ $t \in \mathbb{R}$ $\quad x = 5 + 4t$, $y = 2 - 2t$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 + 4t \\ 2 - 2t \\ t \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 4 \\ -2 \\ 1 \end{bmatrix}$$

(b)

$$\begin{aligned} x \quad\quad\quad\quad &= 1 \\ y \quad\quad\quad &= 2 \\ z - 3w &= 4 \end{aligned}$$

5.

$\boxed{x = 1}$, $\boxed{y = 2}$

$z = 4 + 3w$

let $\boxed{w = t}$ $t \in \mathbb{R}$

$\boxed{z = 4 + 3t}$

$$X = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 4+3t \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 3 \\ 1 \end{bmatrix}$$

$t \in \mathbb{R}$.

(T) 5. Consider the linear system

$$\begin{aligned} x + 2y + 3z &= -1 \\ x - 2y - z &= -1 \\ 3x + y + z &= 3. \end{aligned}$$

(a) Find all solutions, if any exist, by using the Gaussian elimination method.

REF → row echelon form

Sol

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & -1 \\ 1 & -2 & -1 & -1 \\ 3 & 1 & 1 & 3 \end{array}\right]$$

$$\xrightarrow[-3R_1 + R_3 \to R_3]{-R_1 + R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & -1 \\ 0 & -4 & -4 & 0 \\ 0 & -5 & -8 & 6 \end{array}\right]$$

$$\xrightarrow{-\frac{1}{4}R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & -5 & -8 & 6 \end{array}\right]$$

$$\xrightarrow{5R_2 + R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -3 & 6 \end{array}\right]$$

$$\xrightarrow{-\frac{1}{3}R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -2 \end{array}\right] \quad \text{REF}$$

from R_3 → $\boxed{z = -2}$

from R_2 → $y + z = 0 \Rightarrow y = 2$

From R_1

$$x + 2y + 3z = -1$$

$$x + 4 - 6 = -1$$

$$\boxed{x = 1}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$$

T 5. Consider the linear system

$$\begin{aligned} x + 2y + 3z &= -1 \\ x - 2y - z &= -1 \\ 3x + y + z &= 3. \end{aligned}$$

(b) Find all solutions, if any exist, by using the Gauss–Jordan reduction method. → RREF

reduced

Sol

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & -1 \\ 1 & -2 & -1 & -1 \\ 3 & 1 & 1 & 3 \end{array}\right]$$

$$\xrightarrow[-3R_1+R_3 \to R_3]{-R_1+R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & -1 \\ 0 & -4 & -4 & 0 \\ 0 & -5 & -8 & 6 \end{array}\right]$$

$$\xrightarrow{-\frac{1}{4}R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & -5 & -8 & 6 \end{array}\right]$$

$$\xrightarrow{5R_2+R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -3 & 6 \end{array}\right]$$

$$\xrightarrow{-\frac{1}{3}R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -2 \end{array}\right] \text{REF}$$

$$\xrightarrow{-2R_2+R_1 \to R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

$$\xrightarrow[-R_3+R_2 \to R_2]{-R_3+R_1 \to R_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -2 \end{array}\right] \begin{array}{l} \to x \\ \to y \\ \to z \end{array}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$